



Daffodil
International
University

Project Proposal

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Title: ExpressWay Courier Control

Introduction: The purpose of this document is to outline the software requirements for the development of a Courier System Application. The application will utilize a system to assist users in easily sending couriers from one location to another location without any difficulties. We developed a driver side to easily find out the agent's (Our partner who helps to send and receive the parcel) location to pick up and drop off the parcels. The driver also can find the user's location if it is necessary to deliver the parcel to the home location. This will help the user to send the parcel without any difficulties and it makes the courier system totally decentralized. For emergency services, we can send parcels within 24 hours using the logistic support we design.

Problem Identification:

- **Distance and Accessibility:** Users must travel long distances to reach courier hubs, making the service inconvenient.
- **High Costs:** Existing services are often expensive, discouraging users from utilizing them for smaller or urgent shipments.
- **Lack of Decentralization:** Centralized systems limit parcel pickup and drop-off locations, causing delays and inefficiencies.
- **Inadequate Real-Time Tracking:** Users often lack access to real-time shipment tracking, leading to uncertainty about delivery times.
- **Communication Gaps:** Insufficient communication between users, agents, and drivers results in misunderstandings and delays.
- **Limited Service Offerings:** Current platforms do not cater to diverse user needs, such as emergency courier services.
- **Inefficient Route Management:** Non-optimized routes lead to longer delivery times and increased fuel costs for drivers.
- **Fragmented User Experience:** Lack of integrated platforms creates a confusing experience for users managing their courier activities.
- **Inadequate Support for Agents and Drivers:** Agents and drivers lack effective tools to manage orders and communicate with customers.
- **Market Saturation with Limited Differentiation:** Many courier platforms offer similar features, failing to address specific user needs effectively.

Motivation: The motivation behind developing the courier service application is to bridge the gap between the user and the administrator of the courier service system provider, especially in situations When we want to courier a parcel we have to go to Savar or Mirpur. It is a very long distance and time-consuming to me. I have another option to courier from home but it is very costly. One is time-consuming and the other is expensive. There is not any middle point to courier any parcel. Which is not more time-consuming and also not very costly. So we are trying to solve this problem and the decentralization of the courier system. We can easily courier any parcel from any place. We will create a small hub at a minimum distance to easily post and receive a parcel.

Objective: The objective of the Courier system application is to provide a convenient and accessible platform for users to send and receive couriers remotely and conveniently. The application aims to assist in managing the courier system, emergency courier sending, and provide information about nearby agent locations and services.

User Requirements:

- User registration and authentication to create accounts and log in securely.
- Ability to easily place and submit courier orders with sender and recipient details.
- Real-time shipment tracking using a unique tracking ID or order number.
- Option to schedule pickups at user convenience.
- Access to order history for reviewing past shipments.
- Automated notifications via email or SMS for order updates.
- Nearby agent location finder for easier drop-off and pick-up.
- Secure payment processing with multiple options.
- Access to customer support for inquiries and complaints.
- Ability to provide feedback and rate delivery experiences.
- Option for emergency courier services for urgent deliveries.
- Intuitive and responsive user interface for a seamless experience.
- Information on promotions and discounts for shipments.

Features:

Order Management:

- Create and manage courier orders, capturing details for both senders and recipients.
- Assign unique tracking IDs to each order for easy identification and tracking.

Shipment Tracking:

- Enable customers to track their shipments using the provided tracking ID or order number.
- Provide real-time updates on the status and location of shipments.

Route Optimization:

- Utilize algorithms to optimize delivery routes, minimizing delivery times and fuel costs.
- Generate optimal routes for delivery personnel based on order locations and current traffic conditions.

Customer Portal:

- Develop a customer portal for users to place orders, schedule pickups, and view their order history.
- Implement automatic notifications and alerts to keep customers informed about changes in order status.

Rider Portal:

- Create a dedicated portal for delivery personnel to receive order assignments, navigate routes, and update delivery statuses.
- Allow riders to send notifications upon successful deliveries.

Agent Portal:

- Enable agents to create and manage courier orders, including sender and recipient details.
- Provide tracking capabilities for agents to monitor tracking IDs and assist users with support inquiries.

Admin Dashboard:

- Offer administrators a comprehensive dashboard for managing orders, tracking shipments, and monitoring performance metrics for agents, users, and riders.
- Generate reports on delivery efficiency, order volume, and customer satisfaction.

Payment Integration:

- Create invoices for customers based on their order history.
- Include options for both online and offline payment processing.

Notifications and Alerts:

- Send automated email or SMS notifications to customers regarding order confirmations, dispatch updates, and deliveries.
- Notify drivers of new assignments and route changes.

Inventory Management:

- Allow admins and agents to maintain and manage inventory details and stock levels.
- Provide alerts for high inventory levels or items that need to be shipped.

Customer Support:

- Implement a customer support system to address inquiries, complaints, and assistance requests.

Technology Stack:

1. **Front-End**
 - HTML, CSS, JavaScript
 - Responsive Design: Tailwind
2. **Back-End**
 - Programming Languages: PHP
3. **Database**
 - Relational: MySQL
4. **APIs & Integrations**
 - We use our needed API
5. **Payment Processing (Optional)**
 - Secure payment gateways
6. **Version Control & CI/CD**
 - Git/GitHub for version control

Outcome:

- Development of a functional courier service application with multiple functionality.
- Integration with external services and APIs for emergency courier services, location-based agent, and live parcel information facility.
- Enhanced accessibility to courier services and information for users.

Budget and Financial Plan:

1. Initial Development Costs
 - Personnel salaries
 - Technology (licenses, tools, hosting)
 - Design costs
1. Operational Expenses
 - Maintenance (server and database)
 - Customer support (staff and tools)
 - Marketing (campaigns and materials)

Future Scope:

- Development of a functional courier service application with multiple functionality.
- Integration with external services and APIs for emergency courier services, location-based agent, and live parcel information facility.
- Enhanced accessibility to courier services and information for users.

Related Works

These are just a few examples of Courier system platforms that are available in Bangladesh. They're similar to us but not the same in all sections. and we try to add new features to our Courier system platform to fulfill all the requirements of users.

- Pathao Courier is a courier system platform that allows users to express delivery and standard delivery, nationwide delivery, fulfillment solutions, cash-on-home delivery, corporate service, and parcel return.
- RedX offers courier services through its website. The website allows users to same-day, next-day, and on-demand delivery services. Its features include parcel tracking, key account management, proof of delivery, returns management, etc.
- Ecourier is a courier service system that offers home delivery, business parcel delivery, air parcel delivery, and logistical support in Bangladesh. The platform offers a variety of services, including store pickup & return services which enhance customer experience.

If we search online, there are so many courier service platforms but no one can fulfill all the requirements of the patient. So there is a big gap between doctors and patients and we try to fill this gap.

Conclusion: The courier service application aims to provide a convenient and accessible platform for users to send and receive parcels. It will be low cost for all users. Agents can earn extra income and drivers can use their trip regularly and earn more money. From all side it will be a helpful platform.